

Group guidelines. For 3DCRT plans, two or three lateral and oblique fields were used whereas four or five fields were used for IMRT planning. Prescription dose and normal tissues constraints were identical for 3DCRT and IMRT plans. The prescribed dose was 59.4 Gy delivered at 1.8 Gy per fraction using 4-6 MV photons. Normal tissue dose constraints were 50-54 Gy for optic chiasm/nerves and 55-60 Gy for brainstem. Tumor and normal tissue dose volume histograms were generated and compared. Tumor control and normal tissue complication probabilities were also calculated and compared.

**Results:** IMRT resulted in superior target coverage compared to 3DCRT. Specifically, minimum and mean planning target volume doses were 54.52 and 61.73 Gy for IMRT and 50.50 and 60.06 Gy for 3DCRT, respectively ( $p \leq 0.01$ ). In addition, the percent planning target volume receiving below 59.4 Gy was remarkably lower with IMRT (3.30%) compared to 3DCRT (22.76%,  $p \leq 0.0001$ ). IMRT significantly decreased the doses to brainstem, brain and optic chiasm compared to 3DCRT, while no difference was found in the dosing of optic nerves and cochlea. Specifically, IMRT reduced the percent volume of brainstem receiving above 45 Gy by 30.96% ( $p = 0.004$ ) as well as the percent volumes of brain above 18, 24 and 45 Gy by 9.95% ( $p = 0.059$ ), 13.79% ( $p = 0.015$ ) and 40.21% ( $p \leq 0.0001$ ), respectively. The percent volume of optic chiasm receiving above 45 Gy was also reduced with IMRT by 30.40% ( $p = 0.047$ ). Compared to 3DCRT, IMRT significantly increased tumor control probability ( $p \leq 0.005$ ) and lowered normal tissue complication probability for brain and brainstem ( $p < 0.033$ ).

**Conclusions:** IMRT improved target coverage while reducing radiation dose to the brain, brainstem and optic chiasm. Recent advances in cancer imaging may be used in combination with IMRT to allow for more accurate target volume definition and radiation dose intensification without augmenting normal tissue toxicity. Moreover, combining IMRT with new promising systemic agents and targeted therapies may allow us to overcome radioresistance, and, therefore, improve outcomes in patients with high-grade glioma.

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### 1113 Gamma Knife-Based Stereotactic Radiosurgery for Uveal Melanoma

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**Purpose/Objective(s):** To examine the treatment outcomes of Gamma Knife (GK)-based stereotactic radiosurgery (SRS) for uveal melanoma at our institution.

**Materials/Methods:** From 1998 through 2004, nineteen patients with uveal melanoma were treated with GK-based SRS in our department. Out of the 19 patients treated, two had tumors abutting the optic disc, and two had tumors involving the ciliary body; these patients would not have been eligible for eye plaque therapy. The apical height of the tumor ranged from 3 to 13 mm (median 4.4 mm). The maximum basal diameter ranged from 5 to 15 mm (median 9 mm). The median age of the patients was 68 years (range 26-83 years), with a median Karnofsky Performance Status (KPS) of 90 (range 70-100). A Leksell stereotactic headframe was used for immobilization of the head which was turned to the contralateral side during frame placement to bring the target eye to a more central position in the frame. The eye to be treated was anesthetized by supraorbital and infraorbital nerve block and immobilized by suturing the lateral and inferior rectus muscles to the periorbital soft tissue. A high resolution magnetic resonance imaging (MRI) with contrast was performed for the GK procedure, which allowed clear visualization of the ocular melanoma in all 19 patients treated. Plugs were used in selected cases to minimize the lens and the optic nerve doses.

**Results:** The radiation dose was 40 Gy prescribed to the 50% isodose line for all 19 patients. The median target volume was 0.24 cc (range 0.06-1.3 cc). The median treatment volume was 0.68 cc (range 0.26-3.2 cc). All patients were serially followed by their ophthalmologists. The median follow-up time was 40.1 months (range 7.3-76 months). The 3- and 5-year overall survival rates were 86.2% and 54.7%, respectively. The 3- and 5-year tumor control rates were both 94.4%. Out of the 19 patients treated, one developed tumor progression after initial shrinkage, one had stable findings on follow-up ophthalmologic examination, and 17 had tumor shrinkage. The 3- and 5-year free-from-distant-metastasis (FFDM) rates were 72.7% and 64.6%, respectively. Six of the 19 treated patients (32%) developed distant metastasis 30.7 to 74.8 months after SRS. All of them developed liver metastases alone or along with metastases to lungs (2 patients) or bones (1 patient). 5 of the 6 patients (83.3%) had succumbed to metastatic disease at the time of analysis. The apical height of the tumor was a predictor for distant metastases (3 year FFDM rates of 100% vs. 33.3% for patients with tumor apical heights  $< 5$  mm vs.  $> 5$  mm,  $p = 0.017$ ). Out of the 19 patients treated with SRS, 3 had improved, 6 had stable and 10 had worse vision in the treated eye. Two, one and one patients developed vitreous hemorrhage, vitreitis, and conjunctivitis after SRS, respectively.

**Conclusions:** GK-based SRS appears to provide excellent local control of uveal melanoma. It is capable of treating patients not amenable to eye plaque brachytherapy. The risk of distant metastasis is significant, with liver the most common site of involvement. Distant metastasis is the most common cause of death, and the long latent period for its development underscores the importance of long-term follow-up of these patients. Effective systemic therapy is to be explored to improve the treatment outcome of uveal melanoma.

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### 1114 Carbon Ion Therapy of Skull Base Chondrosarcomas

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**Background:** Carbon ions offer physical and biological advantages in the treatment of chondrosarcomas associated with a low radioresponsiveness against photons.